NITTESH KUMAR

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Work Experience

COSMOS Research Center — Little Rock, Arkansas
Software Engineer (Data) II — *July 2025 – Present*

- Architected Cosmos-Dataverse, an in-house scraping platform (React, TypeScript, FastAPI, MongoDB) with role-based access control, using Selenium and Playwright to replace Apify and save $15K/year across 20+ case studies and millions of scraped posts.
- Designed an LLM-based relevancy filter (Gemma 4 31B) with confidence scoring, backed by Kafka job queueing for fault-tolerance.
- Built MethodVerse, a full-stack platform orchestrating 10+ pluggable LLM/ML analysis pipelines, with Qwen3-VL multimodal embeddings (text, image, video) in Qdrant for similarity search and clustering across 1M+ posts.
- Architected database schemas and data models across MongoDB collections for scalable storage; automated backups via Bash scripts, with instant replication for MinIO (S3 compatible open source object storage) on data arrival and scheduled backups for MongoDB.
- Deployed and scaled open-source LLMs (HuggingFace) via vLLM on NVIDIA A40 GPUs, dynamically activating standby nodes under load to sustain thousands of inference requests/minute.
- Co-developed CServe, a FastAPI control plane for LLM serving (vLLM) on a 36-GPU cluster; in production 4+ months, matching bare-vLLM throughput at 4× a Ray Serve baseline.
- Optimized CI/CD pipelines with Jenkins, cutting deployment times by 20%, and managed scalable infrastructure with Docker.
- Developed monitoring dashboards in Grafana/Kibana using Prometheus and ELK, with automated Slack alerting for incidents.
- Built Claude Code workflows using MCP, custom Skills, and subagents to automate implementation, testing, debugging.

L'Enfant Capital Partners — Washington DC
Quant Developer (Contract) — *Aug 2024 – Dec 2024*

- Built a market data pipeline and backtesting engine on AWS EC2/S3, ingesting live equity data via IBKR API and running Monte Carlo simulations across 100+ strategies and 30 assets to evaluate risk-adjusted performance.
- Automated signal generation, trade execution, and logging, delivering end-of-day performance reports to stakeholders.

SIMPLi — Inner Harbor, Maryland
Data Analytics (Contract) — *Jun 2024 – Aug 2024*

- Architected a data lake and warehouse from scratch as a founding engineer, staging raw EDI feeds, distributor reports, and spreadsheets in Azure Data Lake Storage (ADLS Gen2) and building Python/dbt pipelines into Azure Synapse Analytics.
- Built Power BI dashboards tracking 30+ KPIs across sales, marketing, and supply chain teams.
- Developed sales forecasting models to identify high-performing regions and trends.

Techlogix — Karachi, Pakistan
Software Engineer — *Mar 2022 – Jul 2023*

- Built end-to-end test automation frameworks spanning browser (Selenium/Python, Cypress/JavaScript) and API layers (RestAssured/Java), automating 100+ features and validating 50+ backend services, cutting manual testing effort by 40%.
- Designed and developed the data infrastructure for AdalFi on Azure using ADLS Gen2, Key Vault, Microsoft Entra ID with RBAC, and VNet private endpoints, supporting 1.2B+ transactions across 8+ partner banks.

TPS Worldwide (Fintech) — Karachi, Pakistan
Software Engineer — *Aug 2021 – Mar 2022*

- Built backend services for State Bank's RAAST payment gateway, ensuring ISO 20022 compliance.
- Designed RESTful APIs in C++, TypeScript, and Node.js with stream processing, boosting transaction performance 20%.
- Optimized advanced SQL, stored procedures, CTEs, and query tuning, for 15+ banks on TPS IRIS, resolving transaction issues.

Open Source Contribution

Microsoft Azure | *Issue* | *Pull Request*

- Implemented a make aks-delete target in CAPZ to automate **Azure Kubernetes** cluster cleanup and prevent orphaned resources.

Skills

Programming: Python, C++, C, Java, R, HTML, CSS, JavaScript, Bash, PowerShell;
Data & ML: MySQL, NoSQL, Scikit-learn, TensorFlow, PyTorch, Spark, Hadoop, Airflow, dbt, Snowflake, Databricks
Cloud & Devops: AWS, GCP, Docker, Kubernetes, Jenkins, Grafana, Kibana, ElasticSearch, Kafka, Prometheus, CI/CD;
Frameworks & Tools: FastAPI, Node.js, React, Selenium, Cypress, PowerBI, Tableau, MongoDB, Neo4j, REST APIs, Git;

Education

University of Maryland — College Park, United States
Master of Science - Information Systems (Terrapin Scholar) — *Aug 2023 – Dec 2024*

National University of Computer and Emerging Sciences — Karachi, Pakistan
Bachelor of Science - Computer Science — *Aug 2017 – May 2021*

Certifications

- AWS Certified Data Engineer